

Module 3: Data Link Layer and Wireless Networks



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Data-Link Layer

- Data Link Control (DLC)
- Multiple Access Protocols (MAC)
- Link-Layer Addressing
- Ethernet Protocol
- Connecting Devices

Wireless Networks

- Wireless LANs
- Mobile IP

Hands-On

- Datalink Provider Interface
- SOCK_PACKET
- PF_PACKET

Link-Layer Addressing

- IP addresses identify the source and destination hosts in the Internet.
- IP addresses alone cannot determine the path a datagram follows.
- Different datagrams between the same hosts may take different routes.
- Therefore, an additional addressing mechanism is required at each link.
- This mechanism uses **Link-Layer Addresses**.

Also Known As

Link Address, Physical Address, or MAC Address.

Why Are Link-Layer Addresses Needed?

- IP addresses remain unchanged from source to destination.
- Routers forward packets through multiple links.
- Each link requires local source and destination addresses.
- Link-layer addresses identify the two nodes connected by a specific link.
- These addresses change at every hop.

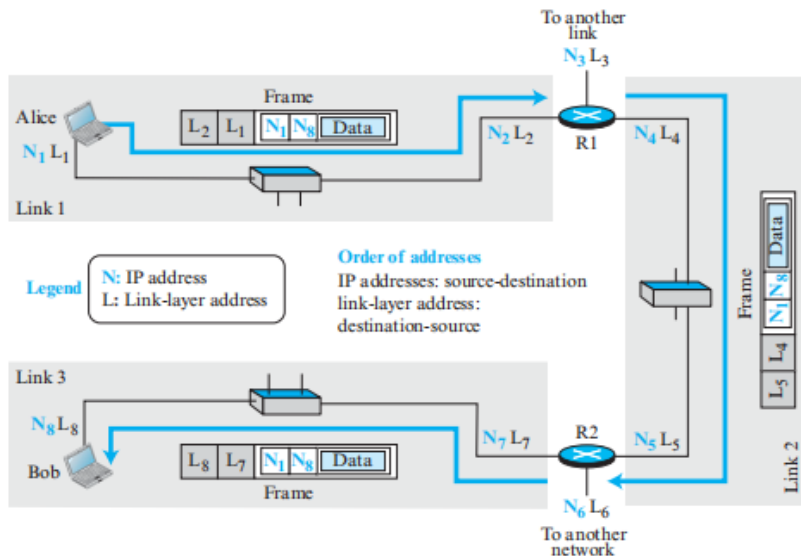
Key Point

IP addresses provide end-to-end delivery, while link-layer addresses provide hop-by-hop delivery.

Encapsulation with Link-Layer Addresses

- A datagram from the network layer is encapsulated into a frame.
- The frame header contains:
 - Destination Link-Layer Address
 - Source Link-Layer Address
- Every router removes the old frame and creates a new frame.
- New link-layer addresses are inserted for the next link.

Figure 5.45 IP addresses and link-layer addresses in a small internet



Address Changes Across Links

Consider communication between Alice and Bob through two routers.

- IP Datagram:
 - Source IP = N1
 - Destination IP = N8
- Frame on Link 1:
 - Destination = L2
 - Source = L1
- Frame on Link 2:
 - Destination = L5
 - Source = L4
- Frame on Link 3:
 - Destination = L8
 - Source = L7



Observation

IP addresses remain constant, but link-layer addresses change at every hop.

IP Address vs Link-Layer Address

Feature	IP Address	Link-Layer Address
Layer	Network Layer	Data-Link Layer
Purpose	End-to-End Delivery	Hop-to-Hop Delivery
Changes During Routing	No	Yes
Scope	Entire Internet	Local Link
Used By	Routers and Hosts	Adjacent Nodes

Why Do Routers Need IP Addresses?

Although router IP addresses do not appear in forwarded datagrams:

- Routers sometimes act as senders and receivers.
- Routing protocols exchange information between routers.
- Routers generate control messages such as ICMP.
- Therefore, routers require their own IP addresses.

Example

Routing protocol messages are exchanged directly between routers using router IP addresses.

Why Multiple IP Addresses in a Router?

- A router connects multiple networks.
- Each interface connects to a different link.
- Every interface requires its own IP address.
- A router with n interfaces needs n IP addresses.

Analogy

A house located at the corner of two streets may have two gate addresses—one for each street.

Determining Source and Destination Addresses

IP Addresses

- Source host knows its own IP address.
- Destination IP address is obtained using DNS.
- Both addresses are inserted into the datagram.

Link-Layer Addresses

- Each device knows its own MAC address.
- Destination MAC address is determined using ARP.
- Addresses are inserted into the frame header.

Size of Link-Layer Addresses

- Link-layer address size depends on the data-link protocol.
- Different network technologies may use different address lengths.
- Ethernet uses 48-bit (6-byte) addresses.

Example Ethernet Address:



A2:34:45:11:92:F1

Note

MAC addresses are typically represented using hexadecimal digits separated by colons.

Summary of Link-Layer Addressing

- Link-layer addresses operate at the Data-Link Layer.
- They provide hop-by-hop communication.
- IP addresses remain unchanged throughout the journey.
- MAC addresses change at every link.
- ARP is used to discover destination MAC addresses.
- Ethernet commonly uses 48-bit MAC addresses.

Key Takeaway

IP addresses identify hosts across the Internet, while link-layer addresses identify devices on a local link.

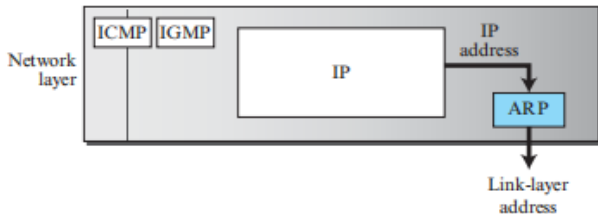
Address Resolution Protocol (ARP)

- IP addresses identify the next node in a path.
- Frames require link-layer (MAC) addresses for delivery.
- ARP maps an IP address to a corresponding MAC address.
- ARP acts as an auxiliary protocol between the Network Layer and Data-Link Layer.

Purpose of ARP

Convert an IP address into a link-layer (MAC) address before frame transmission.

Figure 5.46 *Position of ARP in TCP/IP protocol suite*



Why ARP is Needed

- Hosts know the IP address of the destination or next-hop router.
- Routers know the IP address of the next router from the forwarding table.
- Data-link communication requires MAC addresses instead of IP addresses.
- ARP is used whenever the MAC address of the next node is unknown.

Key Point

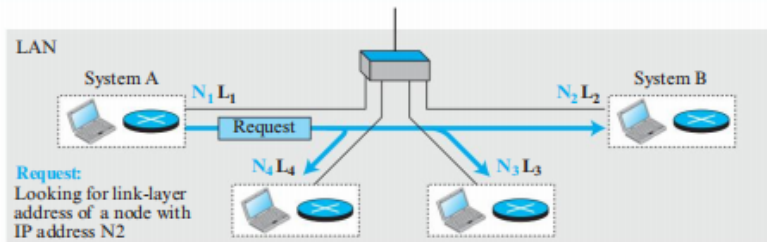
IP addresses identify devices logically, while MAC addresses are needed for actual frame delivery.

ARP Request Process

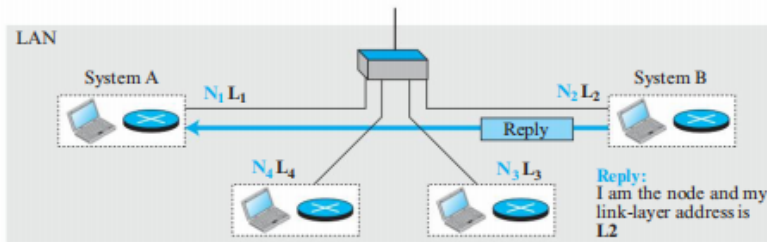
When a sender does not know the destination MAC address:

- ➊ Sender creates an ARP Request packet.
- ➋ Packet contains:
 - Sender IP address
 - Sender MAC address
 - Destination IP address
- ➌ Destination MAC address field is unknown.
- ➍ ARP Request is broadcast to all devices on the local network.

Figure 5.47 ARP operation



a. ARP request is broadcast

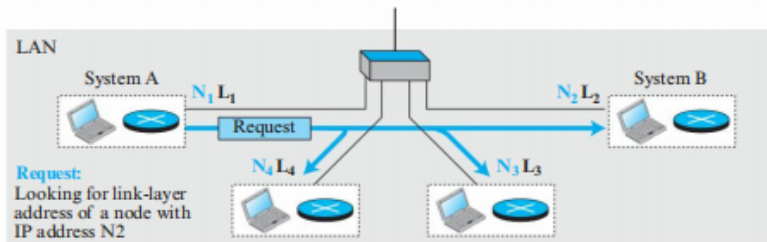


b. ARP reply is unicast

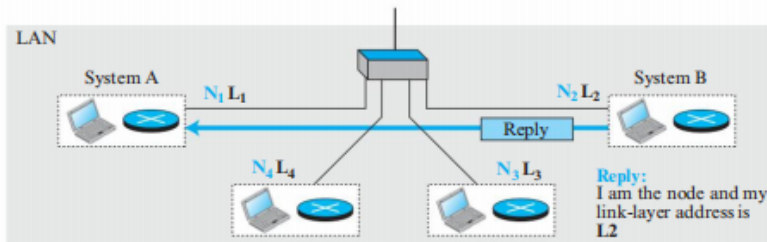
ARP Reply Process

- ① Every host receives the ARP Request.
- ② Only the device whose IP address matches responds.
- ③ Destination sends an ARP Reply packet.
- ④ ARP Reply contains:
 - Destination IP address
 - Destination MAC address
- ⑤ Reply is sent as a unicast message to the requester.

Figure 5.47 ARP operation



a. ARP request is broadcast



b. ARP reply is unicast

Example of ARP Operation

Consider:

- Host A: IP = N1, MAC = L1
- Host B: IP = N2, MAC = L2

Host A wants to send a packet to Host B.

- 1 Host A knows only IP address N2.
- 2 Host A broadcasts an ARP Request asking:

“Who has IP address N2?”

- 3 Host B replies:

“IP address N2 corresponds to MAC address L2.”

- 4 Host A stores L2 and sends frames directly.

ARP Packet Format

ARP packet contains:

- Hardware Type
- Protocol Type
- Hardware Address Length
- Protocol Address Length
- Operation (Request/Reply)
- Source MAC Address
- Source IP Address
- Destination MAC Address
- Destination IP Address

Figure 5.48 ARP packet

0	8	16	31	
Hardware Type		Protocol Type		Hardware: LAN or WAN protocol Protocol: Network-layer protocol
Hardware length	Protocol length	Operation Request:1, Reply:2		
Source hardware address				
Source protocol address				
Destination hardware address (Empty in request)				
Destination protocol address				

Important ARP Fields

Field	Description
Hardware Type	Link-layer protocol (Ethernet = 1)
Protocol Type	Network protocol (IPv4 = 0x0800)
Operation	ARP Request or ARP Reply
Source Hardware Address	Sender MAC Address
Source Protocol Address	Sender IP Address
Destination Hardware Address	Receiver MAC Address
Destination Protocol Address	Receiver IP Address

ARP Request and Reply Frames

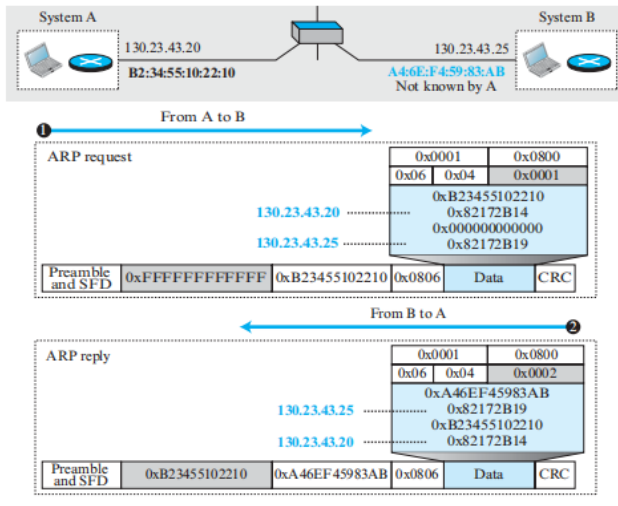
ARP Request

- Destination MAC = Broadcast Address
- Source MAC = Sender MAC
- Contains target IP address

ARP Reply

- Destination MAC = Sender MAC
- Source MAC = Receiver MAC
- Contains target MAC address

Figure 5.49 Example 5.17



ARP Request vs ARP Reply

Feature	ARP Request	ARP Reply
Transmission Type	Broadcast	Unicast
Purpose	Discover MAC Address	Provide MAC Address
Destination MAC Known?	No	Yes
Sent By	Requesting Host	Target Host
Contains Target MAC	No	Yes

Summary of ARP

- ARP maps IP addresses to MAC addresses.
- ARP is required before sending frames on a local link.
- ARP Request is broadcast to all devices.
- ARP Reply is sent only by the intended destination.
- Ethernet commonly uses ARP to resolve MAC addresses.
- ARP enables communication between the Network Layer and Data-Link Layer.

Key Takeaway

Without ARP, a host may know the destination IP address but cannot deliver the frame because the destination MAC address is unknown.

An Example: Addressing Across TCP/IP Layers

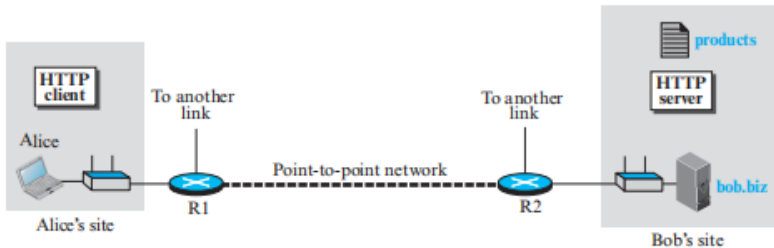
- Physical Layer has no addressing mechanism.
- Data-Link, Network, Transport, and Application layers use different identifiers.
- A user only needs to know the URL of the destination.
- All other addresses are automatically generated by the system.

Example

Alice wants to access:

`http://bob.biz/product`

Figure 5.50 *The internet for our example*



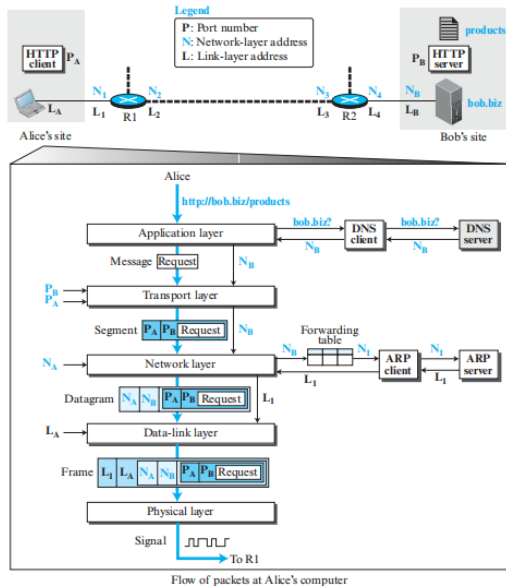
Addresses Used in Different Layers

Layer	Address Used
Application Layer	URL (bob.biz/product)
Transport Layer	Port Numbers
Network Layer	IP Addresses
Data-Link Layer	MAC Addresses
Physical Layer	No Addressing

Key Idea

Each layer adds its own identifier to enable communication.

Figure 5.51 Flow of packets at Alice's computer



Step 1: User Enters URL

Alice enters the URL:

`http://bob.biz/product`

- 1 Browser invokes the HTTP client.
- 2 HTTP client needs Bob's IP address.
- 3 HTTP client requests help from DNS.
- 4 DNS server resolves:

$\text{bob.biz} \rightarrow NB\text{bob.biz} \rightarrow NB_{\text{bob.biz}} \rightarrow NB_{\text{bob.biz}} \rightarrow NB$

Result

Destination IP address NB is obtained.

Step 2: Transport Layer Processing

The Transport Layer creates a TCP segment.

- Source Port = PA (temporary port)
- Destination Port = PB = 80 (HTTP)
- Segment contains HTTP request data.



Segment = (PA , $PB = 80$, $Data$)

Purpose

Port numbers identify the communicating processes.

Step 3: Network Layer Processing

- Network Layer receives destination IP address NB.
- Routing table is consulted.
- Default router R1 is selected.
- Router IP address obtained:



Observation

The destination IP address remains NB, but the packet is forwarded to the next hop router.

Step 4: ARP Resolves MAC Address

- Alice knows router R1's IP address ($N1$).
- Alice does not know router R1's MAC address.
- ARP is used to find:



- MAC address $L1$ is returned.

Result

Network Layer now has the next-hop MAC address.

Step 5: Data-Link Layer Processing

The Data-Link Layer creates a frame.

Addresses Added

- Destination MAC = L1
- Source MAC = LA



Frame = ($L1$, LA , *Datagram*)

Purpose

Frame is ready for transmission on the local network.

Activities at Router R1

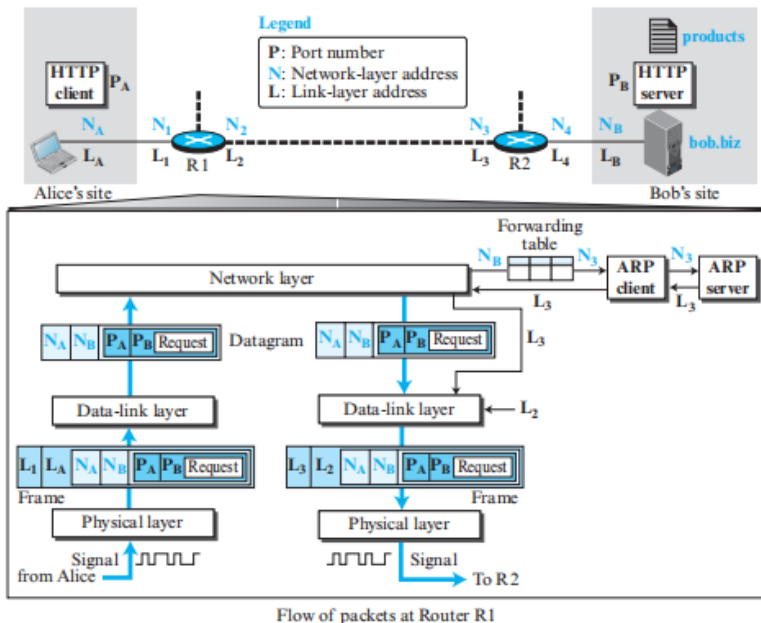
- 1 Frame arrives from Alice.
- 2 Data-Link Layer removes MAC header.
- 3 Network Layer examines destination IP = NB.
- 4 Routing table identifies next router R2.
- 5 Next-hop IP address:

$N3$

- 6 ARP resolves:

$N3 \rightarrow L3$

Figure 5.52 Flow of activities at router R1



Frame Reconstruction at Router R1

Router R1 creates a new frame.

New MAC Addresses

- Destination MAC = L3
- Source MAC = L2



Frame = ($L3$, $L2$, Datagram)

Important

Only MAC addresses change. IP addresses remain unchanged.

Activities at Router R2

Router R2 performs the same operations:

- Receives frame.
- Removes old MAC addresses.
- Checks destination IP address NB.
- Determines next hop.
- Uses ARP if needed.
- Creates a new frame with updated MAC addresses.

Key Observation

Each router changes only the link-layer addresses.

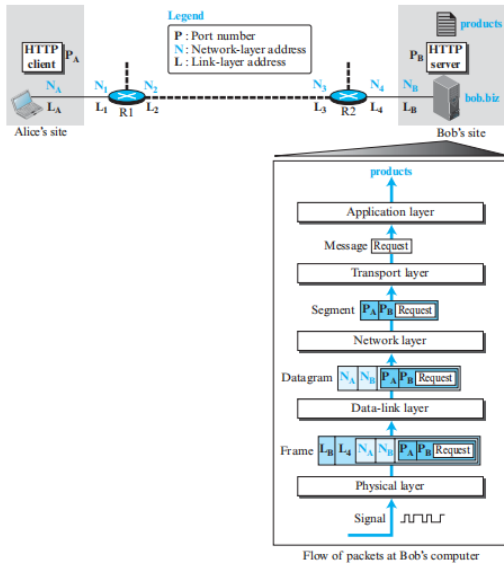
Activities at Bob's Site

- 1 Physical Layer receives signals.
- 2 Data-Link Layer extracts the datagram.
- 3 Network Layer extracts the segment.
- 4 Transport Layer delivers data to port:

$$PB = 80$$

- 5 HTTP server receives the request.

Figure 5.53 Activities at Bob's site



HTTP Server Response

- HTTP server locates the file:

products

- File is formatted as an HTML document.
- Response is sent back to Alice.
- Same protocol stack is used in reverse order.
- Response may travel through routers R2 and R1.

Address Evolution During Communication

Layer	Source	Destination
Application	User Request	URL
Transport	PA	PB = 80
Network	NA	NB
Link 1	LA	L1
Link 2	L2	L3
Link 3	L4	LB

Summary of the Complete Transaction

- User provides only the URL.
- DNS resolves the URL into an IP address.
- Transport Layer creates port numbers.
- Network Layer uses IP addresses.
- ARP resolves MAC addresses.
- Data-Link Layer creates frames.
- Routers update MAC addresses at each hop.
- Destination host decapsulates the message and delivers it to the HTTP server.

Key Takeaway

A complete Internet communication involves URLs, port numbers, IP addresses, and MAC addresses working together across different TCP/IP layers.